

MATH 3330 Homework 8

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Question 1

The last HW asked you to: Prove that if there are simply connected open subsets U and V of X such that $U \cup V = X$ and $U \cap V$ is nonempty and path connected, then X is simply connected. Look through the book's proof of this and make sure you understand it. Note: we are proving a piece of the Seifert-van Kampen Theorem so you can't just say it is true due to Van-Kampen. Why is it important that $U \cap V$ is path connected?

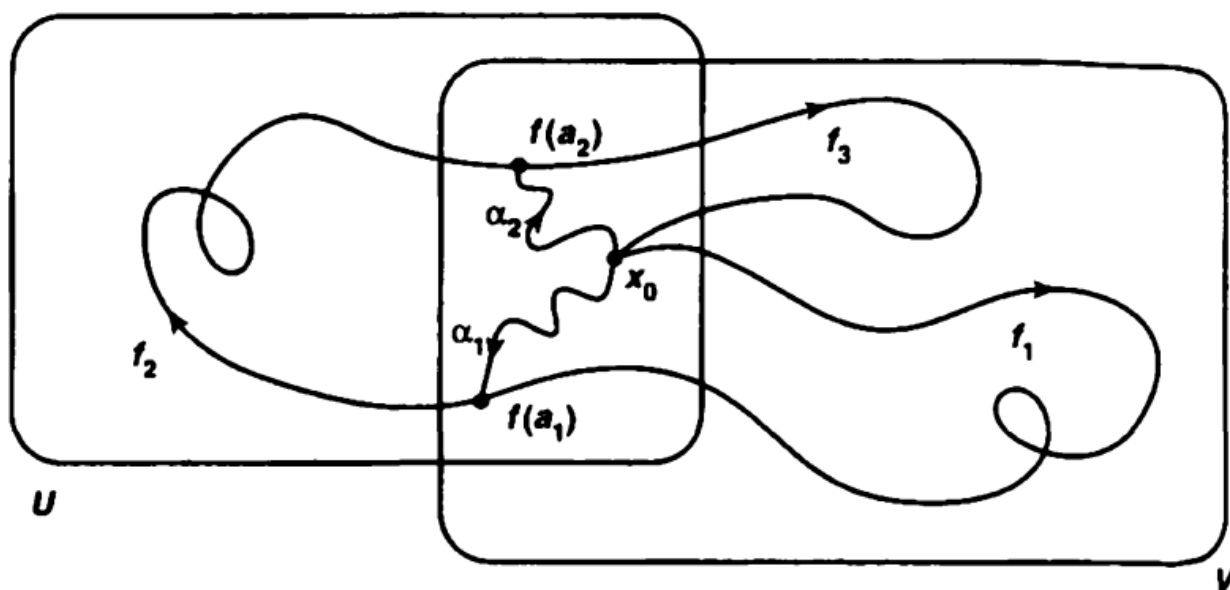


Figure 59.1

$U \cap V$ must be path connected because if it were not, then you could not guarantee that you can connect x_0 to $f(a_n)$ via a path α_n that stays entirely within $U \cap V$. This would then mean that you can't create loops that are entirely contained within U and V separately.

Question 2

For each of the following spaces, the fundamental group is either trivial, isomorphic to $\mathbb{Z}$ or isomorphic to the fundamental group of the figure eight. Determine which one it is in each case.

1. The "solid torus" $B^2 \times S^1$
2. The torus T with a point removed
3. The cylinder $S^1 \times I$
4. $\mathbb{R}^3$ with the non-negative x , y and z -axes removed.
5. $S^1 \cup (\mathbb{R}_+ \times \mathbb{R})$ viewed as a subset of $\mathbb{R}^2$
6. $S^1 \cup (\mathbb{R}_+ \times 0)$ viewed as a subset of $\mathbb{R}^2$

1. The fundamental group of $B^2 \times S^1$ is isomorphic to $\mathbb{Z}$.
2. The fundamental group of a torus T with a single point removed is isomorphic to the figure eight.
3. The fundamental group of $S^1 \times I$ is isomorphic to $\mathbb{Z}$.
4. The fundamental group of $\mathbb{R}^3$ with the non-negative x , y , and z axes removed is isomorphic to the figure eight.
5. $S^1 \cup (\mathbb{R}_+ \times \mathbb{R})$ is isomorphic to $\mathbb{Z}$.
6. $S^1 \cup (\mathbb{R}_+ \times 0)$ is isomorphic to $\mathbb{Z}$.

Question 3

Show that if A is a deformation retract of X and B is a deformation retract of A , then B is a deformation retract of X

Proof: Let A be a deformation retract of X and B be a deformation retract of A . We must show that B is a deformation retract of X .

Since A is a deformation retract of X , then $A \subseteq X$ and there is a map $f : X \rightarrow A$ where $f(a) = a$ for all $a \in A$. The following are true:

1. $f(x, 0) = x$ for all $x \in X$
2. $f(x, 1) \in A$ for all $x \in X$
3. $f(a, t) = a$ for all $a \in A$ and $t \in I$.

Similarly, since B is a deformation retraction of A , $B \subseteq A$ and there is a map $g : A \rightarrow B$ where $g(b) = b$ for all $b \in B$. The following are true:

1. $g(a, 0) = a$ for all $a \in A$
2. $g(a, 1) \in B$ for all $a \in A$
3. $g(b, t) = b$ for all $b \in B$ and $t \in I$.

We can then define the homotopy $h : X \rightarrow B$ as

$$h(x, t) = \begin{cases} f(x, 2t) & t \in [0, \frac{1}{2}] \\ g(f(x, 1), 2t - 1) & t \in [\frac{1}{2}, 1] \end{cases}$$

1. $h(x, 0) = f(x, 2(0)) = f(x, 0) = x$ for all $x \in X$ ✓
2. $h(x, 1) = g(f(x, 1), 2(1) - 1) = g(f(x, 1), 1) \in B$ for all $x \in X$ ✓
3. When $t \in [0, \frac{1}{2}]$, $h(b, t) = f(b, 2t) = b$ and when $t \in [\frac{1}{2}, 1]$,
 $h(b, t) = g(f(b, 1), 2t - 1) = g(b, 2t - 1) = b$ ✓

Since h satisfies each of the above conditions, B must be a deformation retraction of X .

Question 4

Describe the deformation retraction of S^n onto $\mathbb{R}^{n+1} - 0$. Clearly explain why it meets all the conditions of a deformation retraction. [1]

The deformation retraction of $\mathbb{R}^{n+1} - 0$ onto S^n is a function that normalizes all of the vectors beginning at the origin in $\mathbb{R}^{n+1} - 0$. Specifically, we can define a function $f : (\mathbb{R}^{n+1} - 0) \times I \rightarrow S^n$ that projects each point onto the circle. So, $f(x, t) = \frac{x}{(1-t)\|x\|+t}$ where $\|x\|$ is the norm of x . Here $t \in I$ interpolates between x and the projected point on S^n .

We have each of the following:

1. $f(x, 0) = x$ since when $t = 0$ we have not moved.
2. $f(x, 1) = \frac{x}{\|x\|} \in S^n$ is the complete projection onto S^n .
3. $f(a, t) = \frac{a}{(1-t)\|a\|+t} = \frac{a}{(1-t)+t} = \frac{a}{1} = a$.

Therefore this is a deformation retraction.

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1. I think that this is supposed to be the other way, so I am going to solve this question as if it is asking about $\mathbb{R}^{n+1} - 0$ onto S^n ↩